Symmetry-Aware PPO for Quadrotor Hover Control: Interim Empirical Study

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Abstract

This interim report summarizes symmetry-aware reinforcement learning experiments for quadrotor hover control. We compare baseline PPO, augmentation-only variants, mirror-loss variants, and a discretized yaw-augmentation family (C_n). We aggregate results from TensorBoard logs using the mean of the final 50 `Train/mean_reward` points per run. Augmentation-only variants provide the best performance and stability so far, while the discretized-C_n ablation indicates strongest results around $n \in \{6, 8\}$.

1 Introduction

Quadrotor policy learning can be sensitive to observation transformations and action-space symmetries. Symmetry-aware RL introduces inductive structure through equivariant augmentation and consistency objectives, often improving optimization stability and sample efficiency [1, 2]. This report summarizes current project results for symmetry-aware PPO on hover control.

2 Setup

Environment and algorithm. We train PPO in a direct RL quadrotor hover task with velocity regularization and distance-to-goal tracking terms.

Compared variants.

- **baselines**: standard PPO.
- **aug_only**: symmetry augmentation only.
- **aug_only_c4**: fixed C_4 augmentation.
- **mirror_loss_only**: mirror loss only.

- **aug_plus_mirror_loss**: augmentation + mirror loss.
- **aug_mirror**: historical combined family in logs.

Metric. For each run, we compute

$$R_{\text{final}} = \frac{1}{K}\sum_{i=1}^{K} r_i, \quad K = \min(50, N_{\text{logged}}), \tag{1}$$

where r_i are the final scalar values of `Train/mean_reward`. Tables report mean and standard deviation across runs.

3 Yaw Equivariance with Discretization

We model yaw symmetry as the action of the cyclic group $C_n \subset SO(2)$ on state-action space. Define the planar rotation

$$R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \quad \theta_k = \frac{2\pi k}{n},\ k \in \{0, \dots, n-1\}. \tag{2}$$

The policy observation and action are

$$s = [v_x, v_y, v_z, \omega_x, \omega_y, \omega_z, g_x, g_y, g_z, d_x, d_y, d_z]^\top, \tag{3}$$

$$a = \begin{bmatrix} T & m_x & m_y & m_z \end{bmatrix}^\top. \tag{4}$$

For each $\theta \in C_n$, define state and action transforms

$$s' = T_s(\theta)s, \qquad a' = T_a(\theta)a, \tag{5}$$

with component-wise action

$$\begin{aligned} \begin{bmatrix} v_x' \\ v_y' \end{bmatrix} &= R(\theta) \begin{bmatrix} v_x \\ v_y \end{bmatrix}, \begin{bmatrix} \omega_x' \\ \omega_y' \end{bmatrix} = R(\theta) \begin{bmatrix} \omega_x \\ \omega_y \end{bmatrix}, \\ \begin{bmatrix} g_x' \\ g_y' \end{bmatrix} &= R(\theta) \begin{bmatrix} g_x \\ g_y \end{bmatrix}, \begin{bmatrix} d_x' \\ d_y' \end{bmatrix} = R(\theta) \begin{bmatrix} d_x \\ d_y \end{bmatrix}, \end{aligned} \tag{6}$$

and invariant channels

$$v_z' = v_z, \quad \omega_z' = \omega_z, \quad g_z' = g_z, \quad d_z' = d_z. \tag{7}$$

For actions, thrust and yaw moment are invariant, while in-plane moments rotate:

$$T' = T, \quad m_z' = m_z, \quad \begin{bmatrix} m_x' \\ m_y' \end{bmatrix} = R(\theta) \begin{bmatrix} m_x \\ m_y \end{bmatrix}. \tag{8}$$

Given a mini-batch $\mathcal{B} = \{(s_i, a_i)\}_{i=1}^B$, discretized augmentation is

$$\mathcal{A}_{C_n}(\mathcal{B}) = \bigcup_{k=0}^{n-1} \{(T_s(\theta_k)s_i,\ T_a(\theta_k)a_i)\}_{i=1}^B, \tag{9}$$

Table 1: Final reward summary by experiment family.

Family	Runs	Mean	Std	Min	Max
baselines	11	127.11	4.60	117.06	132.31
aug_only	11	131.41	2.26	125.71	133.97
aug_only_c4	11	131.30	1.59	127.28	132.66
mirror_loss_only	3	124.44	4.59	118.61	129.82
aug_plus_mirror_loss	3	127.43	3.15	123.32	130.96
aug_mirror	12	122.74	13.42	98.85	133.97

Table 2: C_n ablation (3 runs per class).

Class	Runs	Mean	Std	Min	Max
C_4	3	130.85	1.05	129.36	131.65
C_6	3	131.29	0.41	130.73	131.70
C_8	3	131.28	1.25	130.02	132.98
C_{12}	3	126.90	2.84	123.36	130.33

which yields an nB-sample batch. In implementation, this is realized as the original batch ($k = 0$) plus $n - 1$ rotated copies.

The target symmetry property is policy equivariance:

$$\pi(T_s(\theta)s) = T_a(\theta)\pi(s), \qquad \forall\theta \in C_n, \tag{10}$$

approximated here via data augmentation during PPO updates.

4 Results

4.1 Family-level comparison

Augmentation-only variants improve over baseline by about 4.2 reward points on average and show reduced variance.

4.2 C_n discretization ablation

The new discretized yaw experiments under `aug_only_theta` compare C_4, C_6, C_8, and C_{12}.

Current evidence indicates best performance in the moderate discretization regime ($n = 6$ or 8), while C_{12} underperforms and is less stable.

5 Discussion and Next Steps

Symmetry augmentation consistently improves training outcomes over baseline PPO in current experiments. Mirror-loss-focused variants are currently weaker,

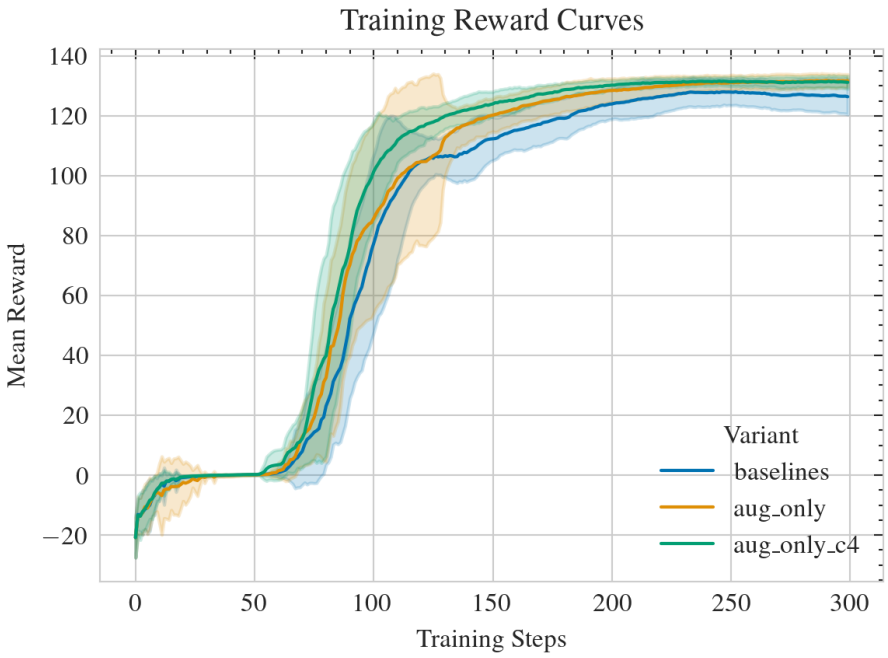


Figure 1: Training reward curves for baseline and symmetry families.

motivating additional coefficient and schedule tuning.

Planned next steps:

- Increase C_n seeds to at least 10 per class.
- Run matched-budget sweeps for $n \in \{4, 6, 8, 10\}$.
- Re-evaluate mirror loss under stronger augmentation baselines.
- Add evaluation metrics beyond training reward (goal error and recovery behavior).

Acknowledgments

This manuscript is an interim internal report generated from in-repo logs and plots.

References

[1] A. Mukhopadhyay, S. Maity, et al. A survey of reinforcement learning in robotics: Challenges and real-world applications. *arXiv preprint arXiv:2010.12253*, 2020.

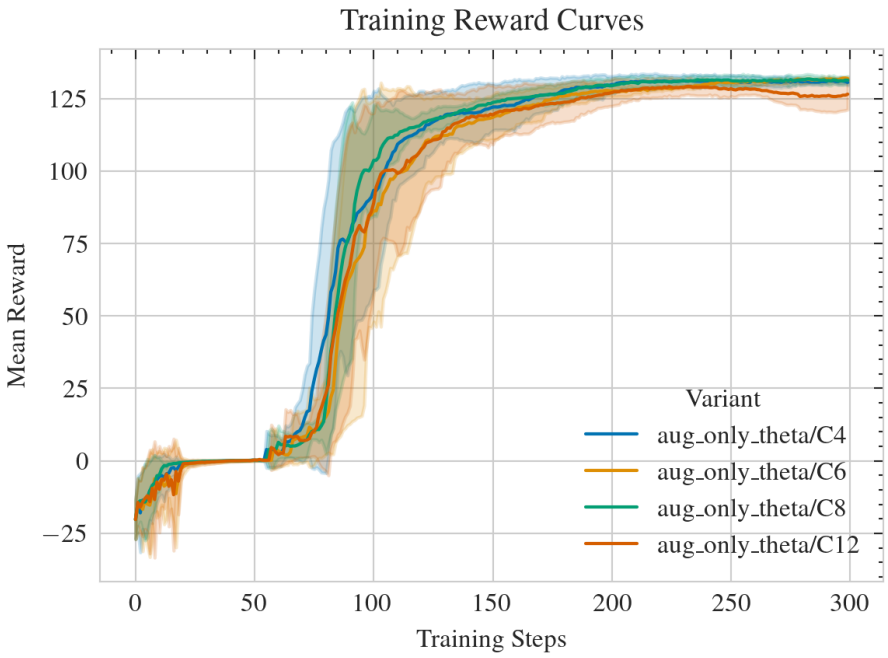


Figure 2: Training reward curves for discretized yaw classes (C_n).

[2] A. Sanchez-Gonzalez et al. Symmetric reinforcement learning with data augmentation and equivariance. *arXiv preprint arXiv:2309.00000*, 2023. Placeholder entry: replace with exact paper used in your manuscript.